CO2 AT 3

COMPILER DESIGN

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1) Scenario : Parsing Table Conflict in LL(1)

In an LL(1) parser, the parsing table is connected using the FIRST and FOLLOW sets of the grammar. The FIRST set determines the terminal that can appear at the beginning.

Reason for Conflict :-

The grammar is not LL(1) because.

* FIRST sets overlap, or
* FIRST and FOLLOW set overlap due to e-productions

Effects on Deterministic Parsing :-

* Parser became non-deterministic
* Multiple production choices exist
* Predictive parsing fails.

Decisions :-

Modify the grammar by:

* Removing left recursion
* Applying left factoring.
* Ensuring FIRST and FOLLOW sets are adjusted

2) ## Scenario : Incorrect Parse Tree Generation :

The parser builds the parser tree according to grammar rules.

For the expression : id + id - id

the tree should represents : (id + id) - id

Influence of Grammar :-

Grammar determines :

* Parse tree structure
* Operator grouping
* Expression evaluation order.

If the grammar is ambiguous, multiple parse tree are possible

Decision :-

Rewrite the grammar to enforce left associativity.

Eg :-

$E \rightarrow E + T$

$E \rightarrow E - T$

$E - T$

or use parser precedence declarations.

3) Scenario : Invalid Expression Handling.

Analysis

Input : a + * b

Tokens are individually valid

id + * id

However, their sequence violates the grammar

Key Issue :-

After the '+' operator, the grammar expects an operand (id). Instead, another operator '*' appears.

Parser Handling :-

Predictive Parser (LL)

* Finds no matching production in the parsing table.
* Reports a syntax error immediately

Shift Reduce Parser (LR)

* Attempts shift / reduce action.
* Finds no valid actions in the parsing table
* Reports syntax error.

4) Scenario : Ambiguity in Conditional Statement :-

Grammar :-

S → if E then S

S → if E then S else S

S → a

Analysis :-

When the parser encounters :

if E then if E then S else S

it cannot decide whether the else belongs to

* the inner if or
* the outer if.

Key Issue :-

This is the Dangling Else Problem.

Impact :-

* Parsing conflicts occur
* LL parsing table contains multiple entries
* Deterministic parsing is impossible

Decision :-

Unmatched → if E then S

5) Scenario :- Operator Precendence Conflict:

Grammar:

$E \to E + E$

$E \to E * E$

$E \to id$

Analysis:

Input: $id + id * id$

The possible tree : $(id + id) * id$

$id + (id * id)$

The correct evaluation is $id + (id * id)$

~~The~~ .

Issue:

The grammar does not specify:

* Operator precedence
* Operator associativity

Applying Precedence

* $*$ has higher precedence than $+$
* Both operators are usually left associative

Decision:

$E \to E + T \mid T$

$T \to T * F \mid F$

$F \to id$